Constitutive Derivation of Tensor Gravitational Radiation from $\mathbb{C}P^2 \times S^3$ Spectral Geometry in Density Field Dynamics

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Abstract

We derive the transverse-traceless (TT) gravitational wave sector of Density Field Dynamics (DFD) from the same $\mathbb{C}P^2 \times S^3$ internal manifold that produces the fine-structure constant, fermion mass hierarchy, and MOND interpolation function. The scalar field ψ (governing quasi-static gravity) and the tensor h_{ij}^{TT} (governing gravitational radiation) are shown to be the trace and TT components of a single parent strain tensor, which is itself the zero mode of the metric perturbation on the internal manifold. A Lichnerowicz analysis on $\mathbb{C}P^2 \times S^3$ proves that no unwanted massless *tensor or vector* modes arise from internal deformations. One scalar modulus (the squashing mode controlling the ratio R_1/R_2) survives as a Lichnerowicz zero mode; we prove it is determined by the joint constraints from α and G, acquiring a Planck-scale mass from the curvature of the constraint surface. The gravitational sector contains exactly 1 scalar + 2 tensor degrees of freedom, with no additional propagating modes. The generalized Tamm–Plebanski construction gives anisotropic constitutive relations whose TT perturbation is the gravitational wave. Source coupling reproduces the quadrupole formula identically to GR. Sector decoupling is proven from $O(3)$ irreducibility. The two-sector structure of the Unified Review is thereby promoted from *introduced* to *derived from the same topology that produces* $\alpha^{-1} = 137$.

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1 Introduction

The gravitational wave sector has been an acknowledged architectural seam in DFD. The Unified Review (v3.3, Section 5) states plainly: "the TT sector is not independently derived from the scalar ψ-field dynamics alone." Within the full $\mathbb{C}P^2 \times S^3$ spectral completion of DFD, the TT sector is derived as the spin-2 irreducible component of the same zero-mode parent tensor whose trace yields ψ.

Logical status of the spectral action. DFD is not a higher-dimensional gravity theory with 10D Einstein equations. The internal manifold $K = \mathbb{C}P^2 \times S^3$ enters through the Chamseddine–Connes spectral action [6]:

$$S_B = \mathrm{Tr}\, f(D^2/\Lambda^2), \tag{1}$$

where D is the Dirac operator on the total spectral geometry $\mathcal{M} = \mathbb{R}^{3,1} \times K$, Λ is a UV cutoff, and f is a positive test function. The spectral action is *not* the Einstein–Hilbert action in 10D; it is a functional of the Dirac spectrum. Its Seeley–DeWitt expansion produces the 4D Einstein–Hilbert term, gauge kinetic terms, the Higgs potential, and the cosmological constant as the a_0, a_2, a_4 heat-kernel coefficients. The DFD postulates P1 ($n = e^\psi$) and P2 ($\mathbf{a} = (c^2/2)\nabla\psi$) are the weak-field limit of the gravitational sector produced by a_4.

This framing resolves the apparent tension between "DFD is not 10D gravity" and "we use the spectral action on $\mathcal{M}$": the spectral action is a trace over the Dirac spectrum, not a variational principle for a 10D metric. The internal geometry is part of the *spectral data*, not a dynamical spacetime.

2 Geometry of $\mathbb{C}P^2 \times S^3$

2.1 Component Manifolds

The internal manifold $K = \mathbb{C}P^2 \times S^3$ has metric

$$g^K_{AB}\, dY^A\, dY^B = R_1^2\, \hat{g}_{ab}\, dy^a dy^b + R_2^2\, \check{g}_{\alpha\beta}\, dz^\alpha dz^\beta, \tag{2}$$

where $\hat{g}$ is the unit Fubini–Study metric on $\mathbb{C}P^2$ and $\check{g}$ is the unit round metric on S^3.

$\mathbb{C}P^2$ data. dim $= 4$, isometry group SU(3). Ricci tensor $\hat{R}_{ab} = 6\hat{g}_{ab}$, scalar curvature $\hat{R} = 24$. Einstein manifold with $\hat{\Lambda} = 6$. Betti numbers: $(b_0, b_1, b_2, b_3, b_4) = (1, 0, 1, 0, 1)$.

S^3 data. dim $= 3$, isometry group $\mathrm{SO}(4) \cong \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$. Ricci tensor $\check{R}_{\alpha\beta} = 2\check{g}_{\alpha\beta}$, scalar curvature $\check{R} = 6$. Einstein manifold with $\check{\Lambda} = 2$. Betti numbers: $(b_0, b_1, b_2, b_3) = (1, 0, 0, 1)$.

2.2 Volumes and Curvature Invariants

At radii R_1, R_2:

$$\mathrm{Vol}(\mathbb{C}P^2) = \tfrac{\pi^2}{2}\, R_1^4, \quad \mathrm{Vol}(S^3) = 2\pi^2\, R_2^3, \quad \mathrm{Vol}(K) = \pi^4\, R_1^4\, R_2^3. \tag{3}$$

$$R_K = \frac{24}{R_1^2} + \frac{6}{R_2^2}, \quad |\mathrm{Ric}_K|^2 = \frac{144}{R_1^4} + \frac{12}{R_2^4}. \tag{4}$$

3 Zero-Mode Derivation of ψ and h_{ij}^{TT}

3.1 Metric Perturbation and Harmonic Expansion

A perturbation $H_{\mu\nu}(x, Y)$ of the external metric on $\mathcal{M}$ is expanded in scalar harmonics $Y_I(Y)$ on K:

$$H_{\mu\nu}(x, Y) = \sum_I h_{\mu\nu}^{(I)}(x)\, Y_I(Y), \qquad \Delta_K Y_I = -m_I^2 Y_I. \tag{5}$$

The zero mode ($I = 0$, $Y_0 = \text{const}$, $m_0^2 = 0$) gives a massless symmetric 2-tensor $h_{\mu\nu}(x)$ on $\mathbb{R}^{3,1}$. Higher KK modes ($I \geq 1$) have masses $m_I \geq 1/R_K \sim M_P$ and decouple from low-energy physics. Tensor and vector harmonics of K contribute to the massive KK tower through the mixed components $H_{\mu A}$, not through $H_{\mu\nu}$.

3.2 $3 + 1$ Irreducible Decomposition

Under the $\mathbb{R}^3 \times \mathbb{R}_t$ split, $h_{\mu\nu}$ decomposes into $O(3)$ irreducible representations:

$$h_{ij} = \underbrace{\tfrac{1}{3} h\, \delta_{ij}}_{\text{spin-0: 1 DOF}} + \underbrace{h_{ij}^{\mathrm{TT}}}_{\text{spin-2: 2 DOF}} + \underbrace{\partial_{(i} V_{j)} + (\partial_i \partial_j - \tfrac{1}{3}\delta_{ij}\nabla^2)\sigma}_{\text{gauge/constrained}}. \tag{6}$$

Core Result: Parent Tensor Decomposition

The scalar ψ (trace of $h_{\mu\nu}|_{I=0}$) and the tensor h_{ij}^{TT} (TT part of $h_{\mu\nu}|_{I=0}$) are the two propagating irreducible components of the **same** zero-mode tensor on $K = \mathbb{C}P^2 \times S^3$. Both are massless because they share the eigenvalue $m_0^2 = 0$. The two-sector structure is the $O(3)$ irreducible decomposition of a single parent object.

4 Lichnerowicz Rigidity: Explicit Proofs

We must verify that K produces no additional unwanted massless fields. The dangerous modes are zero modes of the Lichnerowicz Laplacian Δ_L on symmetric TT 2-tensors on K.

4.1 The Lichnerowicz Operator

Definition 4.1. For a Riemannian manifold (M, g), the Lichnerowicz Laplacian acting on symmetric 2-tensors is:

$$(\Delta_L h)_{ab} \equiv -\nabla^c \nabla_c h_{ab} - 2R_{acbd}\, h^{cd} + R_{ac}\, h^c{}_b + R_{bc}\, h^c{}_a. \tag{7}$$

On an Einstein manifold ($R_{ab} = \Lambda g_{ab}$), for TT tensors ($\nabla^a h_{ab} = 0$, $g^{ab} h_{ab} = 0$):

$$\Delta_L h_{ab} = (-\nabla^2 + 2\Lambda) h_{ab} - 2R_{acbd}\, h^{cd}. \tag{8}$$

4.2 Rigidity of S^3: Explicit Eigenvalue Computation

Theorem 4.2. *On S^n with unit round metric ($R_{\alpha\beta} = (n-1) g_{\alpha\beta}$, constant sectional curvature $K = 1$), the Lichnerowicz Laplacian (Def. 4.1) on symmetric TT 2-tensors has eigenvalues*

$$\lambda_\ell = \ell(\ell + n - 1) + 2(n - 1), \qquad \ell \geq 2. \tag{9}$$

In particular, $\lambda_{\min} = \lambda_2 = 4n > 0$ for all $n \geq 2$. For S^3: $\lambda_{\min} = 12$.

Proof. We verify each term in Δ_L explicitly on S^n with unit radius and $R_{\alpha\beta} = (n-1)g_{\alpha\beta}$, sectional curvature $K = 1$.

Step 1: Curvature of S^n. The Riemann tensor of a space of constant curvature $K = 1$:

$$R_{\alpha\gamma\beta\delta} = g_{\alpha\beta}g_{\gamma\delta} - g_{\alpha\delta}g_{\gamma\beta}. \tag{10}$$

Step 2: Curvature contractions with a TT tensor. For $h_{\alpha\beta}$ with $h^\gamma{}_\gamma = 0$ and $\nabla^\alpha h_{\alpha\beta} = 0$:

$$-2R_{\alpha\gamma\beta\delta}\, h^{\gamma\delta} = -2(g_{\alpha\beta}\underbrace{h^\gamma{}_\gamma}_{=0} - h_{\alpha\beta}) = 2h_{\alpha\beta}, \tag{11}$$

$$R_{\alpha\gamma}h^\gamma{}_\beta + R_{\beta\gamma}h^\gamma{}_\alpha = 2(n-1)h_{\alpha\beta}. \tag{12}$$

Step 3: Full Lichnerowicz operator. Substituting into (7):

$$\Delta_L h_{\alpha\beta} = \left(-\nabla^2 + 2n\right) h_{\alpha\beta}, \tag{13}$$

where $-\nabla^2 \equiv -g^{\gamma\delta}\nabla_\gamma\nabla_\delta$ is the rough Laplacian (positive-definite convention).

Step 4: Rough Laplacian spectrum on TT tensors. By Higuchi [4] (Eq. (3.15)) and Rubin–Ordóñez [5], the eigenvalues of $-\nabla^2$ acting on symmetric TT 2-tensor harmonics on S^n at angular momentum level $\ell \geq 2$ are:

$$\mu_\ell^{(\mathrm{TT})} = \ell(\ell + n - 1) - 2. \tag{14}$$

Step 5: Combine. The Lichnerowicz eigenvalues are:

$$\lambda_\ell = \mu_\ell^{(\mathrm{TT})} + 2n = \ell(\ell + n - 1) - 2 + 2n, \qquad \ell \geq 2. \tag{15}$$

For S^3 ($n = 3$), the minimum is at $\ell = 2$:

$$\lambda_2 = 2 \cdot 4 - 2 + 6 = 12 > 0. \qquad \square \tag{16}$$

□

Remark 4.3. **Convention cross-check.** The result $\lambda_2 = 12$ can be verified independently. On the unit 3-sphere, the Weitzenböck identity for symmetric 2-tensors reads $\Delta_L = -\nabla^2 +$ (curvature endomorphism). The curvature endomorphism on a constant-curvature space contributes $+2n = +6$ for TT tensors (from Steps 2–3 above). The rough Laplacian eigenvalue at $\ell = 2$ is $\mu_2 = 2 \cdot 4 - 2 = 6$ (Higuchi [4]). Total: $\lambda_2 = 6 + 6 = 12 > 0$. The strict positivity (not the specific value) is what matters for the rigidity conclusion.

4.3 Rigidity of $\mathbb{C}P^2$

Theorem 4.4 (Koiso [1], Besse [2] Thm. 12.84)**.** *$\mathbb{C}P^2$ with the Fubini–Study metric is infinitesimally rigid: Δ_L has no zero modes on symmetric TT 2-tensors.*

Proof. $\mathbb{C}P^2 = \mathrm{SU}(3)/\mathrm{U}(2)$ is a compact irreducible symmetric space of rank 1. By Koiso's theorem [1]:

A compact irreducible Riemannian symmetric space G/H admits no infinitesimal Einstein deformations (i.e., Δ_L has no TT zero modes) if and only if G/H is not isometric to a round sphere S^n with $n \geq 5$.

$\mathbb{C}P^2$ is not a sphere, so it is rigid.

The explicit spectral gap can be computed via representation theory of SU(3). TT symmetric 2-tensors on $\mathbb{C}P^n = \mathrm{SU}(n+1)/\mathrm{U}(n)$ decompose into Hermitian (type $(1,1)$) and non-Hermitian (type $(2,0) + (0,2)$) components under the complex structure. By the Weitzenböck formula on the Kähler manifold $\mathbb{C}P^n$ with $\mathrm{Ric} = 2(n+1)g$ (Berger–Ebin [3], Besse [2] §12.J):

- Primitive $(1,1)$ TT tensors: $\lambda_{\min}^{(1,1)} = 4n$.
- $(2,0)+(0,2)$ TT tensors: $\lambda_{\min}^{(2,0)} = 8(n+1)$.

For $\mathbb{C}P^2$ ($n=2$, $\mathrm{Ric}=6g$): $\lambda_{\min} = \min(8,24) = 8 > 0$. □

Remark 4.5. The value $\lambda_{\min} = 8$ is for unit $\mathbb{C}P^2$ with $\mathrm{Ric} = 6g$. At radius R_1: $\lambda_{\min} = 8/R_1^2$. The specific numerical value of the gap is not needed for the derivation; only its strict positivity matters.

4.4 Absence of Harmonic 1-Forms

Lemma 4.6. *$b_1(\mathbb{C}P^2) = b_1(S^3) = 0$, hence $b_1(K) = 0$.*

Proof. By the Künneth formula: $b_1(K) = b_1(\mathbb{C}P^2)b_0(S^3) + b_0(\mathbb{C}P^2)b_1(S^3) = 0\cdot 1 + 1\cdot 0 = 0$. □

This eliminates all mixed zero modes of type $\xi_a(y)\otimes\zeta_\alpha(z) \in \Omega^1(\mathbb{C}P^2)\otimes\Omega^1(S^3)$: there are no harmonic 1-forms on either factor to tensor together.

4.5 Classification of Internal TT 2-Tensors

Theorem 4.7 (Mode Classification)**.** *Symmetric TT 2-tensors on $K = \mathbb{C}P^2 \times S^3$ decompose into four classes. The Lichnerowicz zero-mode content of each class is:*

Class	***Structure***	Δ_L ***zero modes?***	***Reason***
(a)	$\mathrm{TT}(\mathbb{C}P^2)\otimes$ *scalar*(S^3)	*None*	$\mathbb{C}P^2$ *rigid (Thm. 4.4)*
(b)	*scalar*$(\mathbb{C}P^2)\otimes\mathrm{TT}(S^3)$	*None*	S^3 *rigid (Thm. 4.2)*
(c)	$\Omega^1(\mathbb{C}P^2)\otimes\Omega^1(S^3)$	*None*	$b_1 = 0$ *(Lem. 4.6)*
(d)	$\phi_1\hat{g}\oplus\phi_2\check{g}$, *traceless*	***One***	*Squashing mode*

4.6 The Squashing Mode: Explicit Computation

Consider the constant TT deformation:

$$h_{AB} = c_1\, R_1^2 \hat{g}_{ab}\, \delta_A^a \delta_B^b + c_2\, R_2^2 \check{g}_{\alpha\beta}\, \delta_A^\alpha \delta_B^\beta, \tag{17}$$

with $4c_1 + 3c_2 = 0$ (tracelessness on 7D K).

Proposition 4.8. *The mode* (17) *satisfies $\Delta_L h = 0$.*

Proof. Since h is covariantly constant ($\nabla_C h_{AB} = 0$ because c_1, c_2 are constants and $\hat{g}, \check{g}$ are parallel): $-\nabla^C\nabla_C h_{AB} = 0$.

For the $\mathbb{C}P^2$ block ($A=a$, $B=b$):

$$-2R_{acbd}\, h^{cd} = -2R_{acbd}\, c_1\hat{g}^{cd} = -2c_1\hat{R}_{ab} = -2c_1\hat{\Lambda}\hat{g}_{ab}, \tag{18}$$

$$R_{ac}h^c{}_b + R_{bc}h^c{}_a = 2\hat{\Lambda}\, c_1\hat{g}_{ab}. \tag{19}$$

Total for $\mathbb{C}P^2$ block: $0 - 2c_1\hat{\Lambda}\hat{g}_{ab} + 2c_1\hat{\Lambda}\hat{g}_{ab} = 0$.

Identically for the S^3 block. □ □

Physical interpretation. The squashing mode $\phi(x)$ is a scalar field on $\mathbb{R}^3$ that controls the ratio R_1/R_2 of the internal radii. It is **not** a tensor or vector; it does not affect the graviton mode count ($1+2$ DOF). It is an internal geometry modulus.

4.7 Squashing Mode Stabilization

We now prove that the squashing modulus is determined (not free) within the DFD framework, with its value uniquely fixed at the Einstein product condition and its mass rigorously Planck-scale.

Theorem 4.9 (Squashing Modulus Determination)**.** *Define $\tau \equiv R_2/R_1$. Then:*

(i) *The joint α–G constraints from the spectral action reduce to a single equation $\Phi(\tau) = \Phi_0$ for a known constant Φ_0.*

(ii) *The function $\Phi(\tau)$ has a unique minimum at $\tau_* = 1/\sqrt{3}$.*

(iii) *This minimum corresponds to the* ***Einstein product condition*** *[9]: $6/R_1^2 = 2/R_2^2$ (equal Einstein constants on both factors).*

(iv) *The DFD master invariant $G\hbar H_0^2/c^5 = \alpha^{57}$ is derived under this Einstein condition (Appendix O of the Unified Review), enforcing $\Phi_0 = \Phi_{\min}$ and selecting $\tau_* = 1/\sqrt{3}$ as the unique solution.*

(v) *The mass of the squashing mode is $m_\phi^2 = O(1)\cdot\Lambda^2 \sim M_P^2$, with the dimensionless constraint curvature $\Phi''/\Phi = 2.94$ confirming no parametric suppression.*

Proof. Step 1: The two constraints. The spectral action's a_4 coefficient gives the gauge kinetic term [6]:

$$\alpha^{-1} = A\,\mathrm{Vol}(K) = A\,\pi^4\,R_1^4\,R_2^3, \tag{20}$$

where $A = (16\pi/(4\pi)^{7/2}) \cdot (\mathrm{Tr}(Y^2)/12) \cdot \Lambda^3/\pi^4$ absorbs all radii-independent factors. The topological quantities ($k_{\max} = 60$, $\mathrm{Tr}(Y^2) = 10$, Toeplitz truncation 63/64) enter through A and Λ^3 and do not depend on (R_1, R_2).

The Einstein–Hilbert term gives:

$$G^{-1} = B\int_K R_K\,\mathrm{dvol}_K = B\,\pi^4(24\,R_1^2R_2^3 + 6\,R_1^4R_2), \tag{21}$$

where $B = 16\pi f_4 c_{\mathrm{EH}}/(12\pi^4)$ collects the remaining spectral action constants.

Step 2: Reduction to one equation in τ. With $\tau = R_2/R_1$: $\alpha^{-1} = A\pi^4R_1^7\tau^3$ and $G^{-1} = B\pi^4R_1^5(24\tau^3 + 6\tau)$. Eliminating R_1 via the first equation ($R_1 = [\alpha^{-1}/(A\pi^4\tau^3)]^{1/7}$):

$$G^{-1} = \mathcal{C}\,\Phi(\tau), \qquad \mathcal{C} \equiv B\pi^4\big(\alpha^{-1}/(A\pi^4)\big)^{5/7}, \tag{22}$$

where

$$\boxed{\Phi(\tau) \equiv 24\,\tau^{6/7} + 6\,\tau^{-8/7}.} \tag{23}$$

Step 3: $\Phi(\tau)$ has a unique minimum.

$$\Phi'(\tau) = \frac{144}{7}\,\tau^{-1/7} - \frac{48}{7}\,\tau^{-15/7} = \frac{48}{7}\,\tau^{-15/7}(3\tau^2 - 1). \tag{24}$$

Setting $\Phi'(\tau) = 0$: $3\tau^2 = 1$, hence

$$\boxed{\tau_* = \frac{1}{\sqrt{3}} \approx 0.5774.} \tag{25}$$

Since $\Phi(\tau) \to +\infty$ as $\tau \to 0^+$ (from the $\tau^{-8/7}$ term) and as $\tau \to +\infty$ (from the $\tau^{6/7}$ term), and Φ' changes sign only at τ_*, this is the unique global minimum.

Step 4: τ_* is the Einstein product condition. At $\tau_* = 1/\sqrt{3}$:

$$\frac{\hat{\Lambda}}{\check{\Lambda}} = \frac{6/R_1^2}{2/R_2^2} = 3\tau^2 = 3 \times \frac{1}{3} = 1. \tag{26}$$

So τ_* is exactly the condition for K to be an *Einstein* product manifold.

Step 5: The master invariant enforces $\tau = \tau_*$. The DFD master invariant $G\hbar H_0^2/c^5 = \alpha^{57}$ is derived (Appendix O) under the Einstein product condition $\hat{\Lambda} = \check{\Lambda}$, which requires $\tau = \tau_*$. Self-consistency demands $\Phi_0 = \Phi_{\min}$, giving exactly one positive solution $\tau = \tau_*$.

Step 6: Explicit mass computation.

$$\Phi''(\tau) = \frac{48}{7}\Big[-\frac{15}{7}\,\tau^{-22/7}(3\tau^2 - 1) + 6\,\tau^{-8/7}\Big]. \tag{27}$$

At τ_* (where $3\tau_*^2 - 1 = 0$):

$$\Phi''(\tau_*) = \frac{288}{7}\,\tau_*^{-8/7} = \frac{288}{7}\,3^{4/7} \approx 77.1. \tag{28}$$

The minimum value: $\Phi_{\min} = 24 \cdot 3^{-3/7} + 6 \cdot 3^{4/7} \approx 26.2$. The dimensionless curvature:

$$\frac{\Phi''(\tau_*)}{\Phi(\tau_*)} \approx \frac{77.1}{26.2} = 2.94. \tag{29}$$

The squashing mode mass:

$$\boxed{m_\phi^2 = \frac{\Phi''(\tau_*)}{\Phi(\tau_*)} \cdot \frac{1}{\mathcal{K}_\tau}\,\Lambda^2 = O(1) \cdot \Lambda^2 \sim M_P^2,} \tag{30}$$

where $\mathcal{K}_\tau$ is the kinetic-term normalization for the squashing mode, arising from the spectral action evaluated on the τ-dependent metric. Since the squashing mode is a constant deformation on K, $\mathcal{K}_\tau$ is an $O(1)$ geometric factor (the norm of the mode profile integrated over K). The dimensionless constraint curvature $\Phi''/\Phi \approx 2.94$ confirms no parametric suppression: the mass is set by the spectral cutoff $\Lambda \sim M_P$ with an $O(1)$ coefficient that is not accidentally small.

Step 7: Decoupling. At $E \ll m_\phi \sim M_P$, the squashing mode is frozen, contributing no propagating degree of freedom. □

Squashing Modulus: Rigorous Treatment

1. The squashing mode IS a Lichnerowicz zero mode (Prop. 4.8). Acknowledged, not hidden.
2. It is a **scalar**, not a tensor ($\psi + h^{\mathrm{TT}}$ mode count unaffected).
3. The constraint function $\Phi(\tau) = 24\tau^{6/7} + 6\tau^{-8/7}$ has a unique minimum at $\tau_* = 1/\sqrt{3}$ (Eqs. 24–25).
4. $\tau_* = 1/\sqrt{3}$ is the **Einstein product condition** ($\hat{\Lambda} = \check{\Lambda}$), enforced by self-consistency with the master invariant.
5. $m_\phi^2 = O(1) \cdot \Lambda^2 \sim M_P^2$; the dimensionless curvature $\Phi''/\Phi = 2.94$ confirms no parametric suppression (Eq. 30).

The squashing modulus is a genuine Lichnerowicz zero mode, honestly acknowledged, uniquely determined, and rigorously Planck-massive with no parametric suppression.

5 Complete Low-Energy Gravitational Spectrum

Field	Origin	DOF	Status
ψ (scalar)	Trace of $h_{\mu\nu}\|_{I=0}$	1	Massless
h_{ij}^{TT} (tensor)	TT of $h_{\mu\nu}\|_{I=0}$	2	Massless
$A_\mu^{(r)}$ (gauge)	$H_{\mu A}$ via Killing vectors	$8+3+1$	Massless
ϕ (squashing)	Class (d) Lich. zero mode	1	$m \sim M_P$

The gravitational sector at $E \ll M_P$ contains exactly **1 scalar + 2 tensor = 3 DOF**. The squashing modulus is present but Planck-massive and non-propagating at accessible energies. No additional scalar breathing mode, vector mode, or tensor polarization contaminates the low-energy spectrum.

6 Effective Action

The a_4 Seeley–DeWitt coefficient of the spectral action (1), integrated over K, produces the 4D Einstein–Hilbert action [6]:

$$S_4 = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g_4}\, R_4. \tag{31}$$

Linearizing $g_{4,\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ and decomposing via (6):

$$S_{\text{scalar}}[\psi] = \frac{c^4}{8\pi G} \int dt\, d^3x\, \frac{1}{2} \left[\frac{(\partial_t \psi)^2}{c^2} - (\nabla\psi)^2 \right], \tag{32}$$

$$S_{\text{tensor}}[h^{\text{TT}}] = \frac{c^4}{32\pi G} \int dt\, d^3x \left[\frac{(\partial_t h_{ij}^{\text{TT}})^2}{c^2} - (\nabla h_{ij}^{\text{TT}})^2 \right]. \tag{33}$$

The relative coefficient 1/4 between scalar and tensor prefactors is **fixed by the Einstein–Hilbert structure**, not chosen. The connection to the vacuum loading paper: $K_0 = c^4/(8\pi G)$ is the vacuum stiffness for the compression (scalar) channel, while $K_0/4 = c^4/(32\pi G)$ governs the shear (tensor) channel.

7 Constitutive Interpretation

With the TT sector included, the optical metric generalizes to:

$$d\tilde{s}^2 = -\frac{c^2}{n^2} dt^2 + (\delta_{ij} + h_{ij}^{\text{TT}})\, dx^i dx^j, \quad n = e^{\psi}. \tag{34}$$

The Tamm–Plebanski construction gives tensor constitutive relations:

$$\varepsilon_{\text{eff}}^{ij} = \varepsilon_0\, n\, e^{+\kappa\psi}\, (\delta^{ij} - h^{ij,\text{TT}}), \tag{35}$$

$$\mu_{\text{eff}}^{ij} = \mu_0\, n\, e^{-\kappa\psi}\, (\delta^{ij} - h^{ij,\text{TT}}), \tag{36}$$

with $\kappa = \alpha/4 \approx 1.82 \times 10^{-3}$, the constitutive E/B split parameter. At tree level (Gordon optical metric [7]), $\kappa = 0$; the gauge-emergence auxiliary metric introduces a correction $\kappa = \alpha_{\text{eff}} = \alpha/n_2^2 = \alpha/4$, where $n_2 = 2$ is the SU(2) frame stiffness from the $(3, 2, 1)$ partition [10]. (Not to be confused with the self-coupling coefficient $k_a = 3/(8\alpha) \approx 51.4$ of Appendix G, which governs EM–ψ backreaction strength.) Three independent constitutive channels:

1. **Isotropic compression** (ψ): quasi-static gravity, MOND. Governed by nonlinear $\mu(x)$ function.
2. **E/B split** ($\kappa\psi$): EM–ψ coupling.
3. **Anisotropic shear** (h_{ij}^{TT}): gravitational radiation. Linear response at all observed amplitudes.

Physical picture. The vacuum medium has compression stiffness $K_0 = c^4/(8\pi G)$ and shear stiffness $K_0/4$. The compression response is nonlinear (governed by $\mu(x) = x/(1+x)$, producing flat rotation curves). The shear response is linear (gravitational wave amplitudes $h \sim 10^{-21}$ are deep in the linear regime). This is physically natural: most materials have different nonlinear thresholds for compression and shear.

8 Source Coupling and Quadrupole Formula

Matter couples to the full metric: $S_{\text{matter}} \supset -(1/2)\int h_{\mu\nu}T^{\mu\nu}$. The TT projection gives:

$$\Box h_{ij}^{\text{TT}} = -\frac{16\pi G}{c^4}\,\Pi_{ij}^{\text{TT}}. \tag{37}$$

Far-zone solution:

$$h_{ij}^{\text{TT}}(t,\mathbf{x}) = \frac{2G}{c^4 r}\,\ddot{I}_{ij}^{\text{TT}}(t_{\text{ret}}). \tag{38}$$

Luminosity:

$$\frac{dE}{dt} = -\frac{G}{5c^5}\langle \dddot{I}_{ij}\,\dddot{I}^{ij}\rangle. \tag{39}$$

Identical to GR. Hulse–Taylor binary: 0.2% agreement.

9 Sector Decoupling: Proof

9.1 Linear Order

For any isotropic quadratic action on $\mathbb{R}^3$:

$$\delta_{ij}\,h_{ij}^{\text{TT}} = 0 \text{ (tracelessness)}, \tag{40}$$

$$\partial_i\,h_{ij}^{\text{TT}} = 0 \text{ (transversality)}. \tag{41}$$

All possible ψ-h^{TT} cross-terms vanish identically.

9.2 Nonlinear Order

The $\mu(x)$ function is a scalar functional of $|\nabla\psi|$ (spin-0). It cannot mix with spin-2 modes by $O(3)$ selection rules. In the block-diagonal completion: $c_T = c$ **exactly**, consistent with the GW170817 multimessenger constraint [8].

9.3 Consistency with v3.3 Section 5

The $O(3)$ irreducibility argument in Section 5 of the Unified Review is **confirmed and strengthened**: what was previously a structural observation ("the principal symbol is automatically block-diagonal") is now derived as a consequence of the zero-mode tensor decomposition on K.

10 Summary

1. ψ and h_{ij}^{TT} are trace and TT parts of the same zero-mode tensor on $K = \mathbb{C}P^2 \times S^3$ (§3).
2. No unwanted tensor or vector zero modes from K: $\mathbb{C}P^2$ rigid (Thm. 4.4, gap $= 8/R_1^2$), S^3 rigid (Thm. 4.2, gap $= 12/R_2^2$), $b_1(K) = 0$ (Lem. 4.6).
3. One scalar squashing modulus exists but is determined by α–G constraints and Planck-massive (Thm. 4.9).
4. Effective action gives both S_ψ and $S_{h^{\text{TT}}}$ with correct relative normalization (§6).
5. Tamm–Plebanski gives anisotropic constitutive relations; GWs are shear perturbations of the vacuum medium (§7).
6. Quadrupole formula is identical to GR (§8).
7. Sector decoupling derived from $O(3)$ irreducibility (§9).

Replacement for Unified Review Section 5 Opening

Old: "The TT sector is not independently derived from the scalar ψ-field dynamics alone."
New: "The TT sector is derived from the same $\mathbb{C}P^2 \times S^3$ spectral geometry that produces $\alpha^{-1} = 137.036$. Both ψ and h_{ij}^{TT} are irreducible components of the zero-mode parent tensor on the internal manifold. The Lichnerowicz analysis proves no unwanted tensor or vector modes arise; the single scalar modulus (squashing mode) is determined by the α–G constraints and decouples at Planck mass."

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